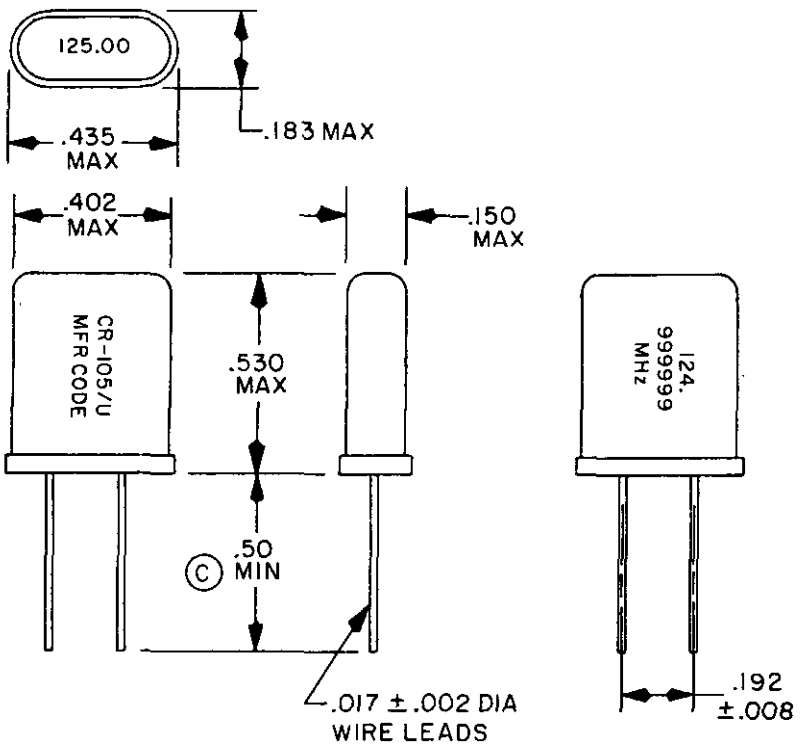
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-105/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 48 to 125 MHz; fifth mechanical overtone; noncontrolled; series resonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-105/U.

© denotes changes

FSC 5955

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:43Z
Check the source to verify that this is the current version before use.

© TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration 1/	See figure 1.
Frequency range	48.0 to 125.0 MHz, inclusive
Capacitance, shunt	4.0 pF maximum
Mode of oscillation	Fifth mechanical overtone
Government reference standard test sets	TS-683/TSM and AN/TSM-15
Rated drive level	2.0 ±0.4 mW
Calibration values:	
Equivalent resistance	40 ohms
Crystal current	0.28 mA
Resonance	Series
Reference temperature:	25°C ±1°C
Frequency tolerance	±10 ppm
Operating temperature range:	-40°C to +95°C, inclusive
Frequency tolerance	±25 ppm
Equivalent resistance	60 ohms, maximum
Unwanted modes	Method II, minimum unwanted-to-main mode resistance ratio of 3:1
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change	±10 percent
Vibration: 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Ordering and marking information: The crystal resonator frequency shall be stamped in two locations on the crystal unit (on top of can as well as its side) in megahertz (MHz) (see figure 1).

2/ Method 204 of MIL-STD-202, test condition A.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

User activities:
Air Force - 17

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0623-15)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:43Z
Check the source to verify that this is the current version before use.